

NONEXISTENCE OF A SINGLE-SAMPLE SELECTION PROCEDURE WHOSE P(CS) IS INDEPENDENT OF THE VARIANCES

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1. Introduction

Suppose that an experimenter has k populations $\pi_1, \dots, \pi_k$, that observations from π_i are normally distributed $N(\mu_i, \sigma_i^2)$ ($1 \leq i \leq k$) with $\mu_1, \dots, \mu_k, \sigma_1^2, \dots, \sigma_k^2$ all unknown, and that the goal is to select a population which has the largest mean, say $\mu_{[k]}$ (where $\mu_{[1]} \leq \dots \leq \mu_{[k]}$ denote the ordered means). Bechhofer (1954) discussed this problem when $\sigma_1^2 = \dots = \sigma_k^2 = \sigma^2$ with σ^2 known; Bechhofer, Dunnett and Sobel (1954) gave a two-sample solution for the case $\sigma_1^2 = \dots = \sigma_k^2 = \sigma^2$ with σ^2 unknown; and various authors considered sequential solutions for these two cases. A satisfactory solution to the problem with $\sigma_1^2, \dots, \sigma_k^2$ unknown has not been found. One reason for this is that certain types of single-sample procedures do not exist. This fact has long been a part of the folklore; e.g., Bechhofer, Dunnett, and Sobel (1954), p. 171, stated that "... a single-sample procedure cannot satisfy the requirements of the problem [probability of a correct selection $P(\text{CS}) \geq P^*$ whenever $\mu_{[k]} - \mu_{[k-1]} \geq \delta^*$, with pre-specified $1/k < P^* < 1$ and $0 < \delta^*$] ..." and that "If the true values of the σ_i^2 are sufficiently large, then it is intuitively clear that the probability of a correct ranking will be arbitrarily close to $1/k$... A rigorous treatment of this point would proceed along the lines used by Dantzig (1940)." In this paper we give a precise statement of this result and three proofs. (Thus, the technique of Bechhofer and Sobel (1954) for the goal of selecting the population with the smallest variance $\sigma_{[1]}^2$, i.e. "washing out" $\mu_1, \dots, \mu_k$ by using sample variances and thereby obtaining a single-sample solution to that problem, has no analogue for the goal of selecting the population with the largest mean $\mu_{[k]}$.)

2. Nonexistence

Let $X_{i1}, \dots, X_{in_i}$ denote $n_i \geq 1$ independent observations from π_i ($1 \leq i \leq k$). One might naturally seek functions such that $g_i = g_{i, n_i}(X_{i1}, \dots, X_{in_i})$ is independent of σ_i^2 ($1 \leq i \leq k$), i.e. such that g_i is a statistic (hence not a function of σ_i^2) and also that the distribution of g_i does not depend on σ_i^2 , and seek to base a selection procedure on them. Let $g_1, \dots, g_k$ be such functions. Let $F_i(x)$ denote the d.f. of g_i . Let $h(x|\xi, \sigma) = (\sqrt{2\pi}\sigma)^{-1} \exp \{-0.5(x-\xi)^2/\sigma^2\}$. Dantzig (1940) proved Theorem 1:

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Theorem 1

If $w \subset \mathcal{R}^m$ is a region such that, for all $\sigma > 0$,

$$(1) \int_w \dots \int_w \left[\prod_{i=1}^m h(x_i | \xi_0, \sigma) \right] dx_1 \dots dx_m = a$$

and

$$(2) \int_w \dots \int_w \left[\prod_{i=1}^m h(x_i | \xi_1, \sigma) \right] dx_1 \dots dx_m = \beta$$

then $a = \beta$.

Theorem 1 allows us to prove

Theorem 2

$F_i(x)$ is independent of μ_i ($1 \leq i \leq k$).

Proof

Recall that g_i is independent of σ_i^2 , so that $F_i(x)$ is independent of σ_i^2 . Thus, when $\mu_i = \mu_0$, $F_i(x)$ is given by (1) where $m = n_i$ and $\xi_0 = \mu_0$. Similarly, when $\mu_i = \mu'_0$, $F_i(x)$ is given by (2) where $m = n_i$ and $\xi_0 = \mu'_0$. In each case $w = \{(y_1, \dots, y_{n_i}) : g_i(y_1, \dots, y_{n_i}) \leq x\}$. The Theorem follows immediately by Theorem 1.

Since our decision is based on $g_1, \dots, g_k$ (which by Theorem 2 are independent of $\mu_1, \dots, \mu_k$ as well as being independent of $\sigma_1^2, \dots, \sigma_k^2$), $P(\text{CS})$ is a constant, say p_k , over that region of $\mathcal{R}^k$ where μ_j (the population mean of π_j) is largest ($1 \leq j \leq k$). (When several populations are tied for largest mean, $P(\text{CS})$ is simply the sum of the appropriate ones of $p_{k1}, \dots, p_{kk}$.) Let $p_k = \min(p_{k1}, \dots, p_{kk})$.

Theorem 3

$p_k \leq 1/k$. Hence $\inf\{P(\text{CS}) : \mu_{[k]} - \mu_{[k-1]} \geq \delta^* \} = p_k \leq 1/k$.

Proof 1

Since $P(\text{CS}) \geq p_k$, we of course have $P(\text{CS}) \geq P^* = p_k$ whenever $\mu_{[k]} - \mu_{[k-1]} \geq \delta^*$; this is achieved with $n_1, \dots, n_k$ observations from $\pi_1, \dots, \pi_k$ (respectively). This procedure implies the existence of a procedure which takes $n = \max(n_1, \dots, n_k)$ observations from each population, ignores the last $n - n_1, \dots, n - n_k$ observations from $\pi_1, \dots, \pi_k$ (respectively), and uses the remaining observations as previously specified. Now Hall (1959) proved that the procedure of Bechhofer (1954) (which required $n'(\sigma) = n'(\sigma, k, P^*, \delta^*)$ observations per population) was most-economical, i.e. that if a single-sample procedure P guaranteed $P_p(\text{CS}) \geq P^*$ (with $1/k < P^* < 1$) whenever $\mu_{[k]} - \mu_{[k-1]} \geq \delta^*$ then it had to require $n_p \geq n'(\sigma)$ observations per population. Thus, $n \geq n'(\sigma)$ and hence $n \geq \sup n'(\sigma) = +\infty$. (This method of proof essentially uses Hall's optimality result in a manner similar to that in which Stein (1945) used a most-powerful-test result to prove Dantzig's theorem.)

Proof 2

$p_{k1} + \dots + p_{kk} = 1$ since: the probability that some population is selected is 1, and the probability that π_i is selected is p_{ki} if π_i is best and is also p_{ki} if π_i is not best (since this probability is independent of $\mu_1, \dots, \mu_k$). Thus $p_k = \min(p_{k1}, \dots, p_{kk}) \leq 1/k$.

Proof 3

$p_{k1} + \dots + p_{kk} = 1$ since when $\mu_1 = \dots = \mu_k$ the $P(\text{CS}) = 1$. But (since $p_{k1}, \dots, p_{kk}$ are independent of $\mu_1, \dots, \mu_k$) this holds for any $\mu_1, \dots, \mu_k$. Thus $p_k \leq 1/k$.

Note that, if the label of the best population does not affect its selection probability, then (since $p_{k1} + \dots + p_{kk} = 1$) $P(\text{CS}) = p_k = 1/k$ (or $rp_k = r/k$ if r populations are tied for largest mean). This is the case if, for example: $g_1, \dots, g_k$ have the same functional form, $n_1 = \dots = n_k = n$ (say), and one selects the population yielding $\max(g_1, \dots, g_k)$ (with randomization over ties).

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